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Intracisternal injection of palmitoylethanolamide inhibits the peripheral nociceptive evoked responses of dorsal horn wide dynamic range neurons

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Abstract Endogenous palmitoylethanolamide (PEA) has a key role in pain modulation. Central or peripheral PEA can reduce nociceptive behavior, but no study has yet reported a descending inhibitory effect on the neuronal nociceptive activity of Aδ- and C-fibers. This study shows that intracisternal PEA inhibits the peripheral nociceptive responses of dorsal horn wide dynamic range cells (i.e., inhibition of Aδ- and C-fibers), an effect blocked by spinal methiothepin. These results suggest that a descending analgesic mechanism mediated by the serotonergic system could be activated by central PEA.

Keywords 5-HT receptors · Analgesia · Descending analgesic mechanisms · Endocannabinoids · Pain

Introduction

As stated by Walker et al. (2002) one of the physiological functions of endocannabinoids is to suppress pain. Accordingly, the palmitoylethanolamide (PEA), the endogenous ligand of PPAR-α (LoVerme et al. 2005a, b), has a well-established role in pain modulation and inflammation (Calignano et al. 1998, 2001; D́Agostino et al. 2007; Genovese et al. 2008; Jaggar et al. 1998; LoVerme et al. 2006). In addition, PEA is able to relieve symptoms of neuropathic pain in humans (Calabrò et al. 2010; Conigliaro et al. 2011; Gatti et al. 2012).

Despite the above findings, no study has yet reported that PEA can exert a descending inhibitory effect on the neuronal nociceptive activity of Aδ- and C-fibers. Indeed, most of these studies have evaluated the peripheral or central effect of PEA on nociceptive behavior, or they have measured molecular markers of inflammation at the spinal cord level (e.g., iNOS, COX-2, NF-κB,). Thus, the present study investigated the effect of intracisternal (i.c.) PEA on the neuronal activity of wide dynamic range (WDR) neurons (at spinal level L4–L5) during peripheral receptive nociceptive electrical field stimulation. Our results show that i.c. PEA inhibits the peripheral evoked nociceptive responses (associated with activation of Aδ- and C-fibers) of WDR cells, suggesting that central PEA could be activating a descending inhibitory pathway. Furthermore, since, the spinal administration of methiothepin (a 5-HT unspecific receptor antagonist) blocks the PEA central effect, a potential descending analgesic mechanism mediated by the serotonergic system is suggested.

Experimental procedures

Animals

Male Wistar rats (260–300 g) were used in the present experiments. The animals were housed in plastic cages in a special temperature-controlled room (22 ± 2 °C) on a 12:12 h light/dark cycle with food and water ad libitum. Our Institutional Ethics Committee approved all animal procedures, and we followed the ethical guidelines for the care and use of laboratory animals established by the

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IASP (Zimmerman 1983) and the NIH (80-23, revised in 1996). All efforts were made to limit distress and to use only the number of animals necessary to produce reliable scientific data.

General methods

Surgical procedures

Thirty-two rats were initially anesthetized in a hermetic box with 3 % sevoflurane in a mixture of two-thirds N_2O and one-third O_2. A cannula was inserted into the trachea to provide artificial ventilation and to maintain the anesthesia throughout the experiment; then, the rats were mounted on a stereotaxic frame. End-tidal CO_2 was monitored with a Capstar-100 CO_2 analyzer (CWI Inc., Ardmore, PA, USA) and kept within physiological range (2.5–3.0 %). Core body temperature was maintained at 38 °C using a circulating water pad. Under these conditions, the animals were secured in a spinal cord unit frame, and lumbar vertebrae were fixed to: (1) perform a laminectomy at spinal cord segments L2–L4 and (2) improve stability at the recording site. In addition, a catheter (PE10) was placed into the cisterna magna for the administration of PEA. The subsequent experimental protocols were performed under 2–2.5 % sevoflurane.

Extracellular unitary recordings

Extracellular unitary recordings were made with: (1) a glass micropipette (impedance 4–8 MΩ) filled with 3 % pontamine sky blue (1 M KCl solution) (21 rats) or (2) seven quartz–platinum/tungsten microelectrodes (impedance 4–7 MΩ) mounted in a multichannel microdrive "System Eckhorn" (Thomas Recording, Giessen, Germany) (12 rats). The microelectrodes were lowered (200–900 μm) into the left laminae of dorsal horn segments. For each recorded cell, the specific somatic receptive field (RF) was located by tapping on the entire ipsilateral hind paw and toes; then, electrical stimulation was applied by two electrodes inserted into the RF (see Electrical stimulation of the somatic receptive field and drug administration section). The extracellular neuronal activity was amplified, digitalized, and discriminated, using CED hardware and Spike 2 software (Cambridge Electronic Design; Version 5.16). The activity of the recorded cells was analyzed with post-stimulus time histograms to identify action potentials with response latencies corresponding to activation of Aβ-fibers (0–20 ms), Aδ-fibers (20–90 ms), and C-fibers (90–350 ms) (Fig. 1). Thus, the number of action potentials that occurred in response to 120 RF stimuli was compared before (basal) and after spinal methiothepin and/or i.c. PEA.

Experimental protocols

Electrical stimulation of the somatic receptive field and drug administration

Two fine needles (27 G) attached to a stimulus isolator unit were inserted subcutaneously into the RF of the recorded neuron. Then, electrical test stimulation was performed; this test consisted of 120 stimuli at 0.5 Hz, with 1-ms pulse duration at 1.5 times the threshold (0.1-3-3 mA) intensity required to evoke a C-fiber response.

We recorded the electrophysiological responses of 39 WDR neurons (mean depth 543.9 ± 40.8) in the basal situation (four consecutive tests with a variation no more than 5 %). Then, when the activity of recorded cells returned to baseline levels, these neurons were divided into seven groups. In one group of cells ($n = 6$, control group, mean depth 715.8 ± 240.0) the effect of the temporal course on the neuronal responses induced by electrical stimulation of the RF (120 stimuli) was evaluated at 0, 10, 20, 40, and 60 min. In four groups (mean depth 420.6 ± 47.2), the effects of i.c. injection of vehicle (DMSO 10 %; $n = 5$; 10 μl) or PEA (1, 3, and 10 μg/10 μl; $n = 5$, 5, and 6, respectively) were analyzed on the neuronal responses induced by electrical stimulation of the RF. The change of neuronal activity induced by PEA (or vehicle) was evaluated at 0, 10, 20, 40, and 60 min.

Finally, two groups of WDR neurons ($n = 6$ each, mean depth 620.8 ± 210.6 and 635.7 ± 192.3, respectively) were used to evaluate the potential serotonergic descending control on the PEA-induced inhibition of the nociceptive evoked responses. In this case, one group was treated only with 15 μg/15 μl methiothepin, a 5-HT unspecific receptor antagonist (control group), whereas the second group was pretreated with 15 μg/15 μl methiothepin (a 5-HT unspecific receptor antagonist) 10 min before the i.c. injection of PEA (10 μg/10 μl). In both cases, the change of neuronal activity was evaluated as describe above for the other groups.

Histology and reconstruction of recording sites

The position of the recorded cells was determined by iontophoretic injection of pontamine sky blue (cathodic current 10–15 μA, 30 min). The animals received an overdose of anesthesia (80 mg/kg of pentobarbital), and were immediately perfused transcardially with saline solution followed by 10 % formalin (~200 ml of each). Brains were postfixed in 10 % formalin and cut in serial coronal sections of 40-μm thicknesses with a Leica SM 200

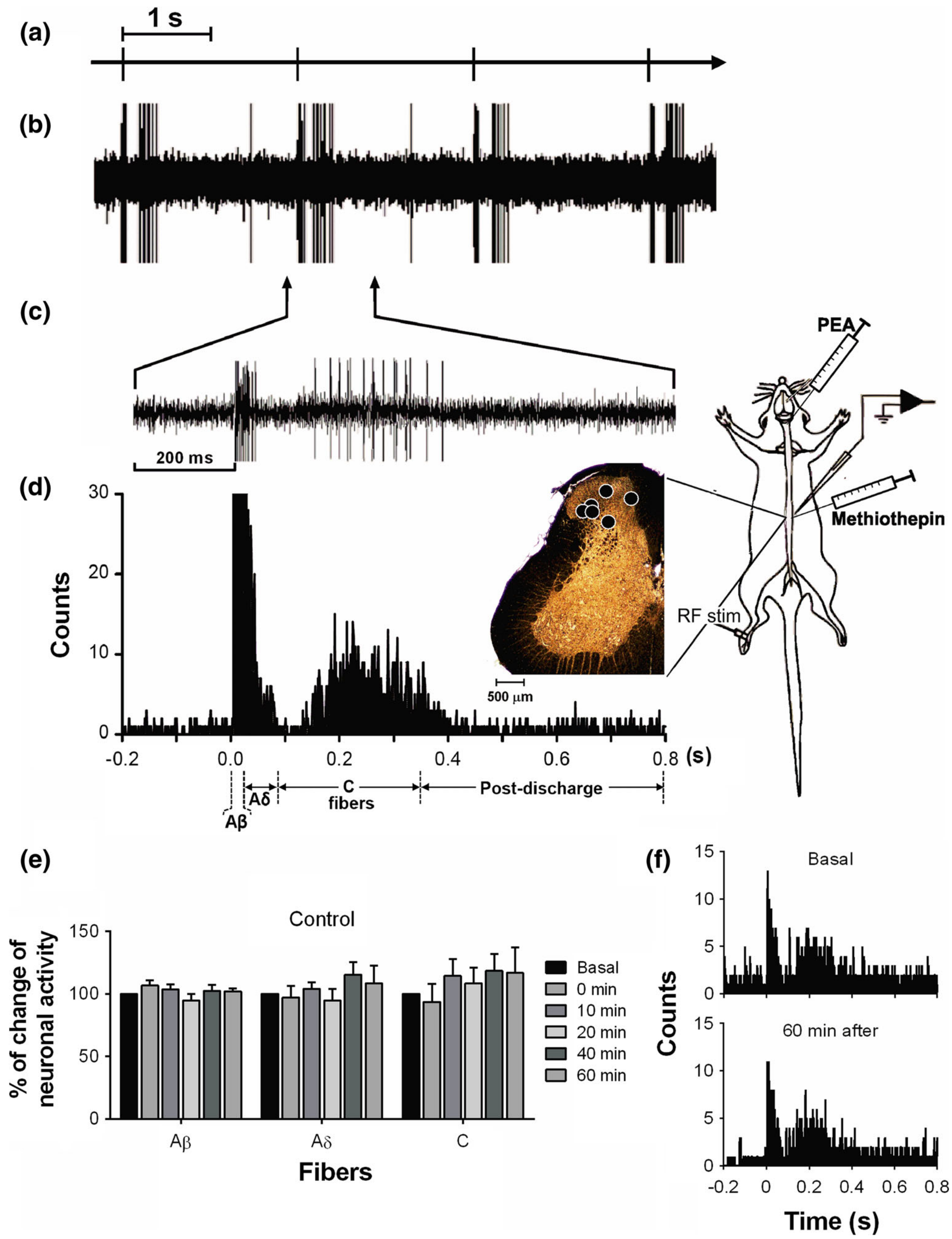

Fig. 1 Neuronal responses to receptive field electrical stimulation (RFst). **a** stimulus artifact induced by the electrical stimulation; **b** four consecutive responses to RFst in an identified dorsal horn wide dynamic range (WDR) neuron; **c** a single response to RFst; **d** the peri-stimulus time histogram obtained from the 120 responses to RFst depicting the different fiber components (Aβ-fibers, Aδ-fibers, and C-fibers), and also it shows a diagram of the experimental design and the location of some recorded cells; and **e** experimental set-up showing the neuronal RF stimulation, the recording site in the lumbar spinal cord, and the histological location of recorded cells on the left side of a spinal cord section. *Panel* **e**, **f** show that the electrical stimulation of the receptive field did not have any effect on the percent of change of neuronal activity during the experiment

R freezing microtome (Leica Instruments GmbH, Nussloch, Germany).

Data presentation and statistical evaluation

Data are presented as mean ± SEM. The changes in neuronal activity are expressed as percent change from the basal response (control). The differences were compared with a one-way repeated measures analysis of variance followed, if applicable, by the Student–Newman–Keuls' post hoc test. Statistical significance was accepted at $P < 0.05$.

Results

General properties of recorded cells

The neuronal responses evoked by the RF electrical stimulation are illustrated in Fig. 1. Extracellular recordings were performed on WDR neurons located in the deep layers of the dorsal horn (laminae IV–V) lumbar enlargement (Fig. 1). All the neurons responded to both innocuous and noxious stimulation of their cutaneous RF and gave responses to Aβ-, Aδ- and C-fiber inputs (Fig. 1c, d) following percutaneous electrical stimulation. Furthermore,

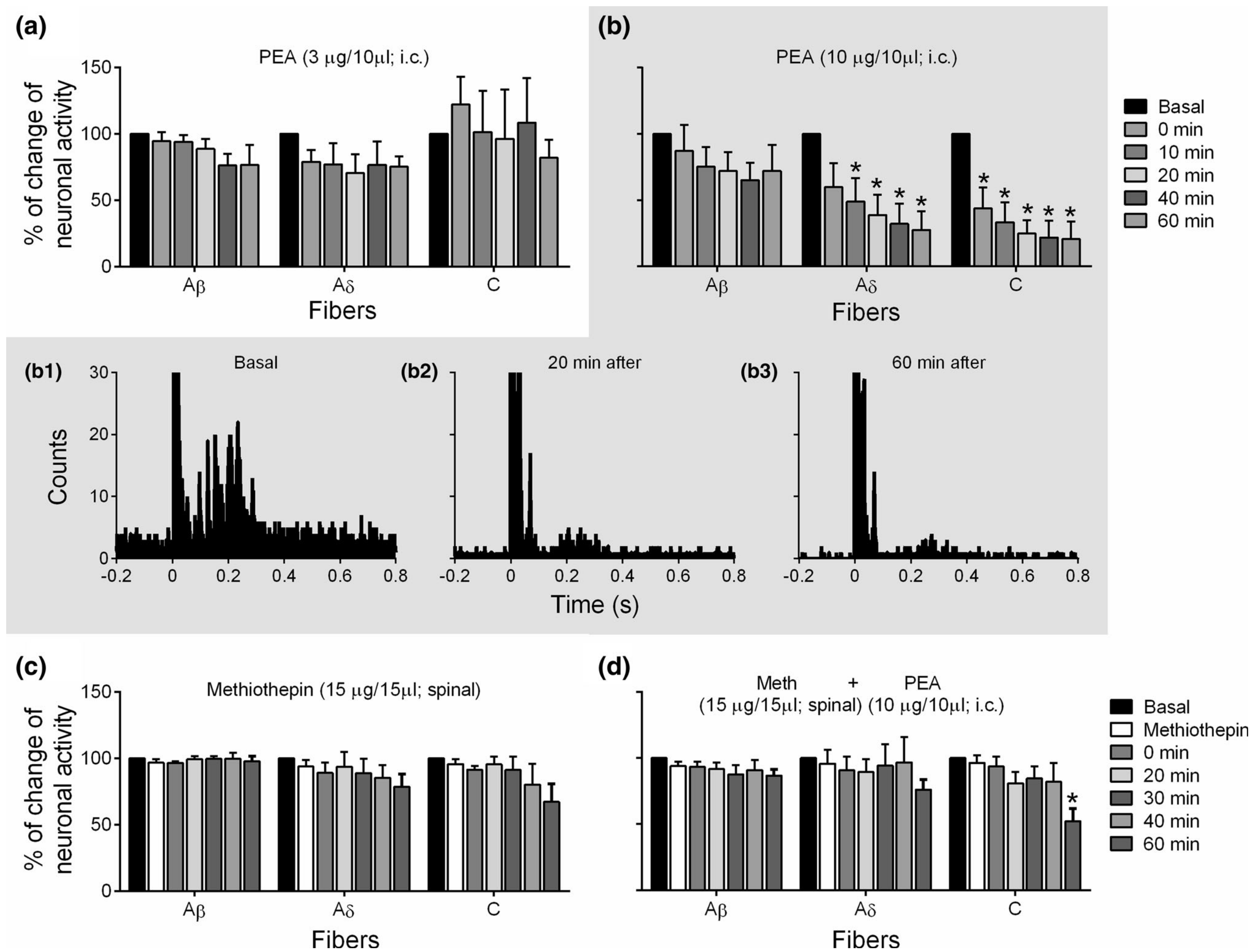

Fig. 2 Effect of intracisternal (i.c.) injection of 3 or 10 µg/10 µl PEA (**a**, **b**, respectively) on the neuronal activity of WDR cells during peripheral receptive electrical field stimulation. Note that only 10 µg/10 µl PEA selectively inhibits the activity of Aδ- and C-fibers but not Aβ-fibers. Firing rate histograms of the same WDR neuron recorded before (control; *b1*), 20 min (*b2*) and 60 min (*b3*) after i.c. administration of 10 µg/10 µl PEA. It is interesting to notice that the spinal pretreatment with methiothepin (an unspecific 5-HT receptor antagonist) blocks the antinociceptive PEA effect (Fig. 2c, d). $*P < 0.05$ vs. control

the Fig. 1e, f show that no time-dependent changes occurred in the neuronal activity for 60 min. Therefore, no time-dependent changes occurred in the neuronal activity over the course of our experiments.

Effects of the i.c. administration of PEA on the nociceptive evoked responses of dorsal horn WDR neurons

Figure 2 shows the analysis of the neuronal responses to RF stimulation before (control) and after i.c. injection of PEA. Similarly to Fig. 1e, Fig. 2a shows that no time-dependent changes occurred in the neuronal activity of animals treated with 3 µg/10 µl PEA; similar results (not shown) were obtained with 1 µg/10 µl and with vehicle (10 % DMSO).

In contrast, Fig. 2b shows that 10 µg/10 µl PEA is able to reduce the neuronal activity corresponding to nociceptive (i.e., Aδ-, and C-fibers) but not to proprioceptive fibers (i.e., Aβ-fibers). In addition, Figures b1, b2 and b3 show experimental tracings illustrating that the neuronal responses to RF stimulation corresponding to nociceptive activity were clearly inhibited since 20 min after the dose of PEA, and they virtually disappeared after 60 min.

Discussion

Central PEA is known to have a role in inflammation and pain modulation (LoVerme et al. 2005a, b, 2006), but its effects on the neuronal activity of Aδ- and C-fibers remain unclear. Although, at the periphery, the intraplantar administration of PEA suppresses the formalin-induced firing rate of spinal cord nociceptive neurons (LoVerme et al. 2006), our work shows that i.c. PEA produces a similar effect suggesting that in addition to this peripheral effect, PEA can activate a descending inhibitory mechanism. These results agree with those of previous study by D'Agostino et al., 2009 showing that in a model of carrageenan-induced paw edema, i.c. PEA reduced the mechanical hyperalgesia and inflammation but its effects were attributed to inhibition of spinal COX-2 and iNOS (proinflammatory enzymes) expression due to central activation of PPAR-α.

Regarding our experimental protocols, we do not measure directly the activity of nociceptive (Aδ-and C) and non-nociceptive fibers (Aβ); instead, we analyze the neuronal activity of the WDR neurons. These types of neurons are located in the dorsal horn of the spinal cord and receive input from the first-order neurons (Aβ-, Aδ- and C-fibers) (Condés-Lara et al. 2003, 2005; Willis 1988). This allows us, to establish the differential participation of each type of fiber, depending on the neuronal responses evoked (see Fig. 1). Since only nociceptive Aδ- and C-fibers were modified by PEA (Fig. 2), we hypothesize that this response: (1) is highly specific and (2) involves a central mechanism. Taken together, the simplest interpretation of these findings is a supraspinal mechanism that modulates primary afferent fibers. Indeed, peripheral noxious stimuli could activate thalamic and cortical areas that, in turn, are able to activate a descending inhibition (Hammond and Yaksh 1981).

Although not specifically characterized in the present study, many possibilities can be proposed to explain how i.c. PEA inhibits at spinal level the nociceptive activity, and some of the most relevant are mentioned in the following lines. In addition to a potential effect modulating mast cells and microglia (Loría et al. 2008; Esposito et al. 2011; Skaper and Facci 2012), the current molecular mechanism of PEA antinociception is related to activation of central PPAR-α receptors (D'Agostino et al. 2009). Indeed, this activation is related to inhibition of spinal COX-2 and iNOS (D'Agostino et al. 2009) enzymes associated with spinal hyperexcitability (Schmidtko et al. 2009; Willingale et al. 1997). Certainly, the use of PPAR-α agonists would be useful to confirm this hypothesis (the role of a central mechanisms mediating inhibition of peripheral nociception).

Further findings supporting the central PEA effect include the fact that inhibition of the fatty acid amide hydroxylase (FAAH) (hydrolytic enzyme of PEA) in the periaqueductal gray area (PAG) inhibits the activity of the ON cells and enhances the activity of OFF cells in the rostral ventromedial medulla (RVM) induced by peripheral intradermal administration of formalin (de Novellis et al. 2008).

Moreover, PEA may act as an entourage compound for endocannabinoids, i.e., it may enhance their action (Lambert and Di Marzo 1999), and it has been shown that cannabinoids interact with descending serotonergic pathways to produce antinociception (Seyrek et al. 2010). In agreement with these findings, our results show that spinal methiothepin, a non-selective 5-HT$_{1/2/6/7}$ receptor antagonist (Andrade et al. 2014) blocked the antinociceptive effect of i.c. PEA (Fig. 2d). In this case, the fact that methiothepin administration (Fig. 2c) had no effects on the neuronal responses induced by peripheral electrical stimulation suggests that: (1) this compound is devoid of any effect per se on the neuronal transmission in our experimental model; and (2) any effect on central PEA-induced inhibition of nociceptive responses is attributable to a direct interaction of methiothepin with spinal 5-HT receptors. Although at this point we cannot ascertain from our experimental procedures the exact nature of 5-HT receptors implicated, Seyrek et al. (2010) show that cannabinoids interact with descending serotonergic pathways and exert a central antinociceptive effect involving spinal 5-HT$_7$ and 5-HT$_{2A}$ receptors, both related to spinal antinociception (Sasaki et al. 2001; Dogrul et al. 2009). On this basis, it is tempting to suggest that the inhibition by 10 μg/10 μl PEA involves a serotonergic mechanism. Admittedly, further experiments are needed to determine the exact nature of 5-HT receptor types involved in this action.

In conclusion, our results show that i.c. PEA inhibits the nociceptive evoked responses of dorsal horn WDR neurons and suggests an inhibition of the activity of the Aδ- and C-fibers. Taken together, these results suggest a central mechanism exerting descending inhibition.

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Conflict of interest The authors declare that they have no conflict of interest.

References

Andrade R, Barnes NM, Baxter G, Bockaert J, Branchek T, Cohen ML, Dumuis A, Eglen RM, Göthert M, Hamblin M, Hamon M, Hartig PR, Hen R, Herrick-Davis K, Hills R, Hoyer D, Humphrey PA, Latté KP, Maroteaux L, Martin GR, Middlemiss DN, Mylecharane E, Peroutka SJ, Saxena PR, Sleight A, Villalon CM, Yocca F (2014) 5-Hydroxytryptamine receptors. IUPHAR database (IUPHAR-DB). Available at: http://www.iuphar-db.org/DATABASE/FamilyMenuForward?familyId=1. Accessed on 02/05/2014

Calabrò RS, Gervasi G, Marino S, Mondo PN, Bramanti P (2010) Misdiagnosed chronic pelvic pain: pudendal neuralgia responding to a novel use of palmitoylethanolamide. Pain Med 11:781–784

Calignano A, La Rana G, Giuffrida A, Piomelli D (1998) Control of pain initiation by endogenous cannabinoids. Nature 394:277–281

Calignano A, La Rana G, Piomelli D (2001) Antinociceptive activity of the endogenous fatty acid amide, palmitoylethanolamide. Eur J Pharmacol 419:191–198

Condés-Lara M, Marina-González N, Martínez-Lorenzana G, Luis-Delgado O, Freund-Mercier MJ (2003) Actions of oxytocin and interactions with glutamate on spontaneous and evoked dorsal spinal cord neuronal activities. Brain Res 976:75–81

Condés-Lara M, Maie IA, Dickenson AH (2005) Oxytocin actions on afferent evoked spinal cord neuronal activities in neuropathic but not in normal rats. Brain Res 1045:124–133

Conigliaro R, Drago V, Foster PS, Schievano C, Di Marzo V (2011) Use of palmitoylethanolamide in the entrapment neuropathy of the median in the wrist. Minerva Med 102:141–147

D'Agostino G, La Rana G, Russo R, Sasso O, Iacono A, Esposito E, Raso GM, Cuzzocrea S, Lo Verme J, Piomelli D, Meli R, Calignano A (2007) Acute intracerebroventricular administration of palmitoylethanolamide, an endogenous peroxisome proliferator-activated receptor-alpha agonist, modulates carrageenan-induced paw edema in mice. J Pharmacol Exp Ther 322:1137–1143

D'Agostino G, La Rana G, Russo R, Sasso O, Iacono A, Esposito E, Mattace Raso G, Cuzzocrea S, Loverme J, Piomelli D, Meli R, Calignano A (2009) Central administration of palmitoylethanolamide reduces hyperalgesia in mice via inhibition of NF-kappaB nuclear signalling in dorsal root ganglia. Eur J Pharmacol 613:54–59

de Novellis V, Palazzo E, Rossi F, De Petrocellis L, Petrosino S, Guida F, Luongo L, Migliozzi A, Cristino L, Marabese I, Starowicz K, Di Marzo V, Maione S, Endocannabinoid Research Group (2008) The analgesic effect of N-arachidonoyl-serotonin, a FAAH inhibitor and TRPV1 receptor antagonist, associated with changes in rostral ventromedial medulla and locus coeruleus cell activity in rats. Neuropharmacology 55:1105–1113

Dogrul A, Ossipov MH, Porreca F (2009) Differential mediation of descending pain facilitation and inhibition by spinal 5-HT3 and 5-HT7 receptors. Brain Res 1280:52–59

Esposito E, Paterniti I, Mazzon E, Genovese T, Di Paola R, Cuzzocrea S (2011) Effects of palmitoylethanolamide on mast cell mediator release and neurotrophic factors in experimental model of spinal cord injury. Brain Behav Immun 25:1099–1112

Gatti A, Lazzari M, Gianfelice V, Di Paolo A, Sabato E, Sabato AF (2012) Palmitoylethanolamide in the treatment of chronic pain caused by different etiopathogenesis. Pain Med 13:1121–1130

Genovese T, Esposito E, Mazzon E, Di Paola R, Meli R, Bramanti P, Piomelli D, Calignano A, Cuzzocrea S (2008) Effects of palmitoylethanolamide on signaling pathways implicated in the development of spinal cord injury. J Pharmacol Exp Ther 326:12–23

Hammond DL, Yaksh TL (1981) Peripheral and central pathways in pain. Pharmacol Ther 14:459–475

Jaggar SI, Hasnie FS, Sellaturay S, Rice AS (1998) The anti-hyperalgesic actions of the cannabinoid anandamide and the putative CB2 receptor agonist palmitoylethanolamide in visceral and somatic inflammatory pain. Pain 76:189–199

Lambert DM, Di Marzo V (1999) The palmitoylethanolamide and oleamide enigmas: are these two fatty acids cannabimimetic? Curr Med Chem 6:757–773

Loría F, Petrosino S, Mestre L, Spagnolo A, Correa F, Hernangómez M, Marzo Di, Docagne F (2008) Study of the regulation of the endocannabinoid system in a virus model of multiple sclerosis reveals a therapeutic effect of palmitoylethanolamide. Eur J Neurosci 28:633–641

LoVerme J, Fu J, Astarita G, La Rana G, Russo R, Calignano A, Piomelli D (2005a) The nuclear receptor peroxisome proliferator-activated receptor-alpha mediates the anti-inflammatory actions of palmitoylethanolamide. Mol Pharmacol 67:15–29

LoVerme J, La Rana G, Russo R, Calignano A, Piomelli D (2005b) The search for the palmitoylethanolamide receptor. Life Sci 77:1685–1698

LoVerme J, Russo R, La Rana G, Fu J, Farthing J, Mattace-Raso G, Meli R, Hohmann A, Calignano A, Piomelli D (2006) Rapid broad-spectrum analgesia through activation of peroxisome proliferator-activated receptor-alpha. J Pharmacol Exp Ther 319:1051–1061

Sasaki M, Ishizaki K, Obata H, Goto F (2001) Effects of 5-HT2 and 5-HT3 receptors on the modulation of nociceptive transmission in rat spinal cord according to the formalin test. Eur J Pharmacol 424:45–52

Schmidtko A, Tegeder I, Geisslenger G (2009) No no, no pain? The role of nitric oxide and cGMP in spinal pain processing. Trends Neurosci 32:339–346

Seyrek M, Kahraman S, Deveci MS, Yesilyurt O, Dogrul A (2010) Systemic cannabinoids produce CB_1-mediated antinociception by activation of descending serotonergic pathways that act upon spinal $5\text{-}HT_7$ and $5\text{–}HT_{2A}$ receptors. Eur J Pharmacol 649:183–194

Skaper SD, Facci L (2012) Mast cell-glia axis in neuroinflammation and therapeutic potential of the anandamide congener. Philos Trans R Soc Lond B Biol Sci 367:3312–3325

Walker JM, Krey JF, Chu CJ, Huang SM (2002) Endocannabinoids and related fatty acid derivatives in pain modulation. Chem Phys Lipids 121:159–172

Willingale HL, Gardiner NJ, McLymont N, Giblett S, Grubb BD (1997) Prostanoids synthesized by cyclo-oxygenase isoforms in rat spinal cord and their contribution to the development of neuronal hyperexcitability. Br J Pharmacol 122:1593–1604

Zimmermann M (1983) Ethical guidelines for investigation of experimental pain in conscious animals. Pain 16:109–110